

Reproducibility checklist (Complete and submit as a standalone file with your project report)

- * Please fill out the top part of the form first (first page).
- * Please make sure the points in second page are addressed in your report submission
- * Please copy this and replace the ☐ with a ✓ for the items that are addressed in your report/code submission
- * Please complete this checklist, attach it to your final project report as the last page and then submit.

Your team member names (comma separated):

Ian Gotcher, Anuj Sakarda, Joshua Lu, Hai Yen Tran

PROJECT GITHUB LINK: <https://github.com/IanGotcher/MisinfoDetectYaleLLMsProject>

Your project title:

Detection, Contextualization, and Explanation of Misinformation
and Disinformation Using Cost-Effective Large Language Models

AI Use Statement (see below for instructions and examples)

We used Google Gemini 3.1 Pro in the Google Antigravity IDE to write code to implement our GUI application with Python, TKInter, and the Google Gemini API.

Author Contribution Statement (see below for instructions and examples)

Ian Gotcher (Grad) thought of the idea to use LLMs to detect misinformation, wrote and submitted the project proposal, created the GitHub repository, wrote the README, obtained the Gemini API key, coded the application, wrote the statements for the benchmark, manually evaluated the application using said benchmarks, wrote the Methods and Approaches slide of the presentation, presented the Results and Analysis slide, and wrote the Motivation and Proposed Application and System Design sections of the final report.

Anuj Sakarda (UG) helped refine the project proposal and did background research, created the Results and Analysis and Conclusion slides of the presentation, and wrote the Related Work section of the final report.

Joshua Lu (Grad) organized the group collaboration and wrote the Intro, Motivation, and Prior Work slide of the presentation, compared the results against the correct labels, calculated the benchmark summary statistics, analyzed representative failure cases and wrote the Results and Evaluation section of the final report.

Hai Yen Tran (UG) conducted diagnostic analysis of model inaccuracies, researched theoretical and technical frameworks for improved implementations, presented the Methods and Approaches slide, and wrote the Discussion and Future Works section of the final report.

Checklist (Make sure these are addressed in your submission and replace ☐ with a ☒ where applies)

Source Code Accessibility:

- ✓ Provide a link to the source code on github or submit the code repo to us separately if you have privacy concerns.
- ✓ Ensure the code is well-documented and easy to run

Computing Infrastructure:

- ✓ Mention the compute resources used (GPUs or API models you've used)

Dataset Description:

N/A Clearly describe and cite the datasets used, and any preprocessing/modifications done.

Hyperparameters and Tuning Process:

N/A Detail the hyperparameters used and the process for selecting them.

(Example: The model was fine-tuned using a batch size of 16, learning rate of 1e-5, and trained on 1000 steps with 100 steps of learning rate linear warmup with linear decay)

Experimental Results:

- ✓ Clearly define the evaluation metrics and statistical methods used in assessing the model.
- ✓ Present the main set of results in a table, chart or readable format

- ✓ Include comparisons with baseline models and state-of-the-art, where applicable.

AI Use

- ✓ Clear statement of how and where you have used AI in the project
- ✓ Specific parts where AI has helped
- ✓ Prompts that you used included as an auxiliary file or appendix in your report

Good Example 1: We used ChatGPT to help brainstorm project directions, clarify implementation details. We then used Claude Code to help write code and experiments. We used both ChatGPT and Claude Code to help plot bar charts for our raw results. All experimental design decisions, code changes, analysis, and final writing were reviewed and finalized by the team. We used ChatGPT to help with polishing the report write-up and help with grammar.

Good Example 2: We used Codex to assist with implementing preprocessing scripts for the summarization dataset, including tokenization, train/dev/test splitting, and formatting examples for model input. We also used Claude Code to help debug errors in the fine-tuning script and to refactor evaluation code for ROUGE scoring. All prompts, generated code, and edits were reviewed by the team before inclusion in the final repository. We did NOT use AI for the final project report writing extensively with exception of helping rephrasing a few sentences.

Bad example 1: We vibe-coded parts of the project using Claude Code.

Bad example 2: AI was used in Writing, Coding and Experiments.
(This doesn't say how exactly you used AI and in what specific parts)

Author Contribution Statement

- ✓ *Clear statement of what each member in your team has done.*

Good example 1: Alice Smith led dataset collection and preprocessing, including cleaning the raw data and creating the train, validation, and test splits. Bob Chen implemented the baseline and fine-tuned models, ran the main experiments, and organized the code repository. Carol Lee designed the evaluation framework, produced the result tables and figures, and wrote the analysis section of the report. All team members contributed to project planning, reviewed the final report, and approved the final submission.

Good example 2 (mix of grad and undergrad students): Bob Chen (UG) handled dataset preprocessing and exploratory analysis. Carol Lee (UG) implemented the model training pipeline and ran the hyperparameter tuning experiments. Alice Smith (Grad) conducted evaluation, created the result visualizations, and wrote the discussion of errors and limitations. Alice also conducted the literature review. All authors contributed to writing and editing the final report.

Bad example 1: All team members contributed equally to the project.

(This gives no information about each person's role or contribution.)

Bad example 2: Alice did coding, Bob did experiments, and Carol wrote the report.

(This is too general, you need to provide specific details on exactly who did what)